# Mingran Xu

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## Education

**University of California San Diego** Sep 2026 - Jun 2028 (Expected)
*Master of Computer Science* *San Diego, CA*

**University of Washington** Sep 2021 - Jun 2026
*Bachelor of Science in Computer Science (GPA: 3.86 / 4.00)* *Seattle, WA*

## Work Experience

**Mobile Intelligence Lab** Feb 2026 - Jun 2026
*Research Assistant* | Python, PyTorch, Hugging Face Transformers *Seattle, WA*

- Built an end-to-end two-stage response gating pipeline in Python to filter delayed LLM-generated responses by detecting topic shifts and redundant content in multi-turn dialogue.
- Developed reproducible data-processing workflows using Python, JSON/JSONL, and Hugging Face Datasets to transform raw dialogue data into structured datasets for supervised classification.
- Fine-tuned and evaluated Transformer-based cross-encoder models with PyTorch, and implemented a CLI-based inference pipeline that outputs accept/discard decisions with model scores and thresholds.

**Volkswagen Group** Feb 2025 - Jun 2025
*Mobility Technology Intern* | Python, PyTorch, PowerBI *Beijing, China*

- Developed and implemented a Power BI-driven data filtering and visualization framework to process data for ADAS/AD assessment.
- Researched synthetic data applications for training and testing of AI-related autonomous driving solutions.
- Assisted colleagues in R&D projects by conducting scientific literature research.

**Mercedes-Benz** Jun 2024 - Nov 2024
*Software Development Engineering Intern* | Java, Spring Boot, Kafka, PostgreSQL *Beijing, China*

- Implemented data persistence and recovery mechanisms using Spring Boot, Kafka, and PostgreSQL, with scheduled data synchronization to improve cross-system consistency and fault tolerance.
- Developed account-deletion workflows to process requests, reconcile records across systems, and prevent redundant account operations.
- Improved account-deletion monitoring by resolving data inconsistencies and eliminating unnecessary alerts.
- Acquired knowledge of AWS cloud platform structure and engaged in designing cloud platforms.

## Project Experience

**Squares Design Web App** | *React, TypeScript, Node.js, Express.js, REST APIs, JSON, Mocha* Aug, 2025

- Built a full-stack design application enabling users to interactively create and edit square-based designs, manage nested layouts, and save designs using a recursive tree-based data model.
- Implemented client-side editing features in React/TypeScript, including square selection, color updates, split/merge operations, and immutable state updates over a tree-structured square data model.
- Developed backend REST APIs for saving, loading, and managing design files, using JSON serialization and Mocha unit tests to validate recursive square operations and server routes.

**News Summarization and Key Sentence Extraction System** | *Python, PyTorch, Transformers, T5, BERT* May, 2024

- Fine-tuned a T5 summarization model with PyTorch on the MultiNews dataset to generate abstractive summaries for multi-document news articles.
- Built a key-sentence extraction pipeline using all-mpnet-base-v2 embeddings to rank source sentences by semantic similarity to generated summaries.
- Implemented a diversity-aware top-k sentence selection algorithm in Python that dynamically compares candidate sentences and filters redundant selections to produce concise, non-overlapping evidence sentences.

## Technical Skills

**Languages**: Java, TypeScript, JavaScript, Python, SQL, C, C++, MATLAB
**Frameworks & Libraries**: Spring Boot, JSON/JSONL, React, Node.js, Express.js, PyTorch, Hugging Face Transformers
**Databases & Infrastructure**: PostgreSQL, Apache Kafka, REST APIs
**Tools & Platforms**: Git, Linux, AWS, Power BI